Swinburne University of Technology

2020-HS2-ICT90003-Applied Research Methods-H1

Canvas Submission

Assignment 3A: Research Proposal

Performance Analysis of Clustering Techniques for Breast Cancer Detection

---

Submitted By:

Oshani Wickramasingha - 102266765

Keyur Chandpara - 102600457

Manish Surisetti – 102628747

Sravani Pottumuthu – 102428769

Ishan Arun Pant - 102413778

Lecturer:

Pro Yun Yang

Tutor:

Dr. Arif Jubaer

Abstract

One of the major reason for cancer deaths can be considered as breast cancer. Digital pathologists face challenges in getting accurate decisions in the detection of breast cancer from stained pathology images. The objective of this research proposal is to identify the best segmentation algorithm out of three clustering algorithms to detect breast cancer with the aid of four performance parameters. The proposed method consists of several phases including data gathering, image enhancement, image segmentation using three clustering methods, train the model, test the model and finally analyzing the best algorithm with the use of four performance parameters. A comparative analysis will be done where the final results will be illustrated with the aid of graphs. CLAHE enhancement technique will be used to increase the quality of the image. Three clustering techniques use for the research are k-means, fuzzy c-means and hybrid k-means algorithm and four parameters are accuracy, sensitivity, balanced accuracy and F-score. Implementation will be carried out in the Jupyter Notebook IDE with the use of UC Santa Barbara dataset. Research will carried out within three months. Estimated cost is approximately AUD 131,813 and will present with minimum ethical issues. Results of the proposed research will help the potential researchers in finding more accurate segmentation algorithms in detection of breast cancer.

Table of Contents

List of Table

List of Figures

1. Introduction

Breast Cancer, a special kind of cancerous tissues involves in the inside layer of the lobules or milk glands which carries milk (Ataollahi et al. 2015). Both women and men can produce these cells of breast cancer but it is slightly uncommon in men's body. These cells grow rapidly and spread through the entire body. The factors affecting breast cancer are from hereditary, breast feeding, overweight, addiction of tobacco and liquor. The second most leading reason for the deaths among women is breast cancer and the prevention of the breast cancer is still challenging since the growth of the cells is a multi-cell process which involves multiple cell types (Sun et al. 2014).

More than 80% cancer patients in developed countries survive up to average five years due to early detection of breast cancer (Sun et al. 2014). Mammography, a most efficient way for the detection of breast cancer in the initial stage (Juliet & Stacey 2020). Diagnosis of breast cancer can be done by using screening methods like Ultrasound, Mammogram, Magnetic Resonance Imaging (MRI), Biopsy and Computer Tomography (CT) scan. (Aswathy & Jagannath 2020; Anon 2020). Histopathology analysis is also used for image segmentation to visualize the images of structures and nuclei in the tissue clearly with high resolution.

Image Segmentation is one of the method designed to acquire the artefacts found in the image by splitting the image to multiple regions which have the similar attributes in an object (Kobashi, Nyul & Udupa 2016). The key aim of image segmentation is to refine the analysis process to make the findings become more understandable. We find the objects and boundaries of an image with high quality and high-resolution segmented image by using image segmentation process (Amalia, Airlangga & Thohari 2018).

Initially the proposal discusses the problem statement with the justification, research aim and questions. Secondly, literature review identifies, evaluates and then synthesizes the literature related to the segmentation algorithms. Research plan includes the methodology detailing the process of the research, limitations and future work. Ethical considerations and project management planning is discussed in the final section before concluding the future suggestions and study expectations.

2. Research Aims and Questions

2.1 Problem Statement

Breast cancer is the second most leading causes of death in the world, after lung cancer. There are accuracy issues in segmentation techniques although there are many segmentation algorithms presents by the researchers. An accurate segmentation algorithm should be find, out of the segmentation algorithms to detect breast cancer which helps radiologists to treat the patients.

2.2 Justification

According to the statistics of Cancer Council of Australia, 17354 women were detected with breast cancer in 2016 (Anon 2020). By 2018, 2999 women were died because of breast cancer in Australia (Aswathy & Jagannath 2020). As per the statistics of WCRI, new cases of two million in the year 2018 identified and majority of deaths rate of women has been increasing day-by-day (Anon 2020). Moving to the western countries, one in eight women are prone to breast cancer (Saunders, Jassal & Lim 2019). Early detection of breast cancer is curable where it considered to be one of the three most frequent cancers in the world (Harbeck & Gnant 2016). As per the global cancer statistics in 2018, 2,088,849 were detected and 626,679 were dead (Bray et al. 2018). New cancer cases were found as 30% in America where 25% of the women (more than 1.5 million) diagnosed with breast cancer (Sun et al. 2014).

Image processing is important for the analyzation of high complex data where radiologists face issues with processing high volume of data in a short period of time. (Kobashi, Nyul & Udupa 2016). Breast cancer detection has become a key issue for the clinicians since the physical diagnosis of histopathology images is very subjective. Time consuming is one of the major drawback of manual segmentation (Heena, Om & Tripti 2014). According to the mammogram images interpreted by the radiologists has shown a low accuracy of 63% which the results has mislead to treat the patients (Kanungo et al 2015).

2.3 Research Aim

This research is aimed to identify the most accurate and efficient algorithm from three main clustering segmentation techniques Fuzzy C-Means clustering, Hybrid K-Means and K-Means clustering by considering the four performance parameters namely Sensitivity, Accuracy, Balanced Accuracy, F-Score. Parameters and segmentation techniques are described in detail in methodology.

2.4 Research Questions

I. How the four parameters sensitivity, accuracy, balanced accuracy, F score impacts on the accuracy of the clustering algorithms?
II. How Contrast Limited Adaptive Histogram Equalization (CLAHE) method helps to improve the quality of the image?
III. Will data be sufficient for training of the model?
IV. Which clustering method is the best out of three algorithms?

Best algorithm will be find by analyzing with the aid of four performance parameters.

3. Literature Review

In this review, different papers regarding the segmentation algorithms are taken into consideration and analyzed for the breast cancer detection. In conclusion, according to a comparison done with several papers it showed that the k-means algorithm has less accuracy than hybrid k-means.

Dubey, Gupta & Jain used k-means clustering algorithm with different measures of computations such as split method, distance and centroid and identified the best measure with highest accuracy (Dubey, Gupta & Jain 2016). Experiments were carried out with the aid of Wisconsin dataset. K-means clustering algorithm used to study classification of BCW dataset. Computational formulation of integrative clustering with the multi-variant parameters were used to investigate accuracy of dataset. The k-means with variations were applied can classify the BCW dataset with single as well as multiple iterations.

Filipczuk, Kowal & Obuchowicz detected the fribroadenoma which is a benign tumor by using hybrid k-means algorithm (Filipczuk, Kowal & Obuchowicz 2012). The classification rate shows using simple methods which give satisfactory results. But it has some problems regarding generation of improper clusters. When nuclei are represented by very small on the image and merges it provides inappropriate results. So, to overcome this problem hybrid k-means is used where it showed more accurate results.

Detection technique, hybrid K-means was used to detect the tumor features from breast cancer diagnosis (Zheng, Yoon & Lam 2014). It is time-consuming process to extract all the related information for utilization. A Support vector machine and new hybrid of k-means used to diagnose the tumor. Mainly k-means algorithm is used for finding hidden patterns of the benign and malignant tumors. And the SVM is used to classify the different incoming tumors. The 10-fold cross validation shows more accuracy of 97.38%. It was tested on Wisconsin Diagnostic breast cancer (WDBC) Dataset. It also helps in better understanding of different property types of tumors (Zheng, Yoon & Lam 2014).

According to Kaur and Aggrawal, the algorithms regarding the effectiveness of k-means, hybrid k- means are discussed in which the hybrid k-means optimization algorithm combines with the optimization algorithms like CGA, KFA. The regular k-means have many problems like when then converge to local minima rather than optimizing results. The hybrid k-means have some other problems like accuracy and efficiency problems. New Hybrid k-means which is used to solve the existing problems associated with the regular normal k-means. The proposed ABCGA [Adaptive Biogeography Clustering Based Genetic Algorithm] technique is more accurate (Kaur, Aggrawal 2017).

4. Research Plan

4.1 Methodology

The cell structures of a histopathology image is very complex and they are in an unorganized manner causing to detect the cancerous cells hard. In this proposing research, the cancerous cells will be identified automatically with the aid of three machine learning algorithms and will evaluate the performance of each algorithm to find the best algorithm.

Methodology used in this research is experimental. Experimental methodology refers for the investigation in which we test a hypothesis where independent variables are manipulated and dependent variables are measured and controls extraneous variables (Leedy & Ormrod 2015).

Proposed research consists of 8 phases namely data collection, image enhancement, image segmentation, training the model, testing the model, analyzing and representation of results in a graph. Independent variable will be the input image and the dependent variable will be the output image which detects whether it's a cancerous image or not. Below diagram shows the flow of the detection of breast cancer process.

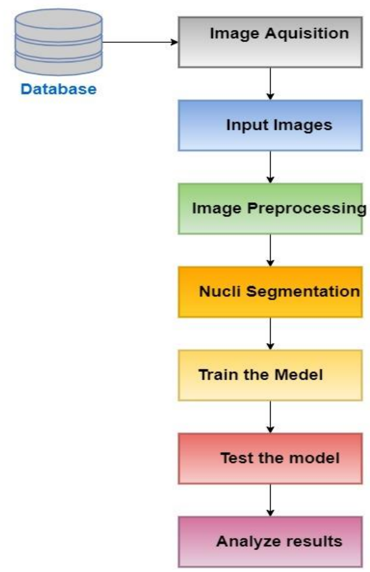
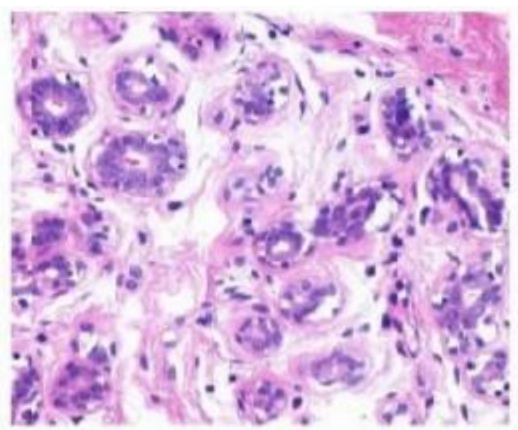


Figure 1: Flow chart of breast cancer detection process

4.1.1 Data Collection

UCSB dataset will be used in the research for the experimentation purpose. Dataset is publicly obtainable in the library of Santa Barbara (The Regents of the University of California 2020). Dataset includes 58 images with a resolution of 896*768 and ground truth images can access in the UCSB library.



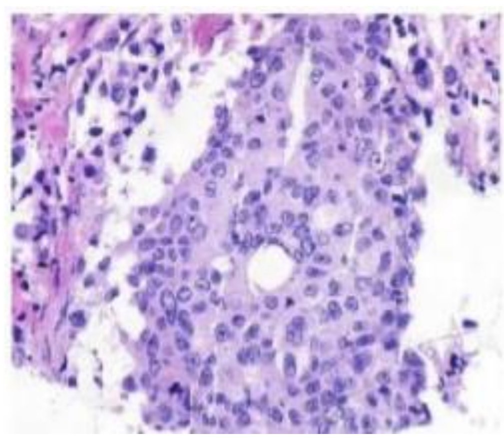


Figure 2: Histopathology image sample which denotes benign case in the left and malignant in the right panel (from UCSB breast cancer cell dataset) (Li & Plataniotis 2015)

4.1.2 Preprocessing

The histopathology images use for the experiment are H&E stained images (Hamatoxylin and Eosin). In the preprocessing stage, the image quality will be enhanced by removing the unnecessary noise where it helps to emerge the malignant information in the image (Kumar, Venkatalakshmi & Karthikeyan 2016). Image enhancement techniques use to enhance the image's visual appearance and to provide a good quality representation for future image processing automation (Janani, Premaladha & Ravichandran 2015). Out of image enhancement techniques such as Imaadjust, histogram equalization and Contrast Limited Equalization (CLAHE), CLAHE technique is used for the image enhancement (Attia, Salim & Hussein 2017).

CLAHE enhances the quality of the image by adjusting saturation and increasing the contrast, value and the hue (Attia, Salim & Hussein 2017). CLAHE focusses on input image's minor patches rather than focusing on the whole image where the contrast of each patch is enhanced separately. In the end, both enhanced image histogram and the histogram which is quantified by the distribution parameters will be matched. In order to combine the patches, bilinear interpolation will be carried out on the neighboring patches. Below figure shows an enhanced image using CLAHE method.

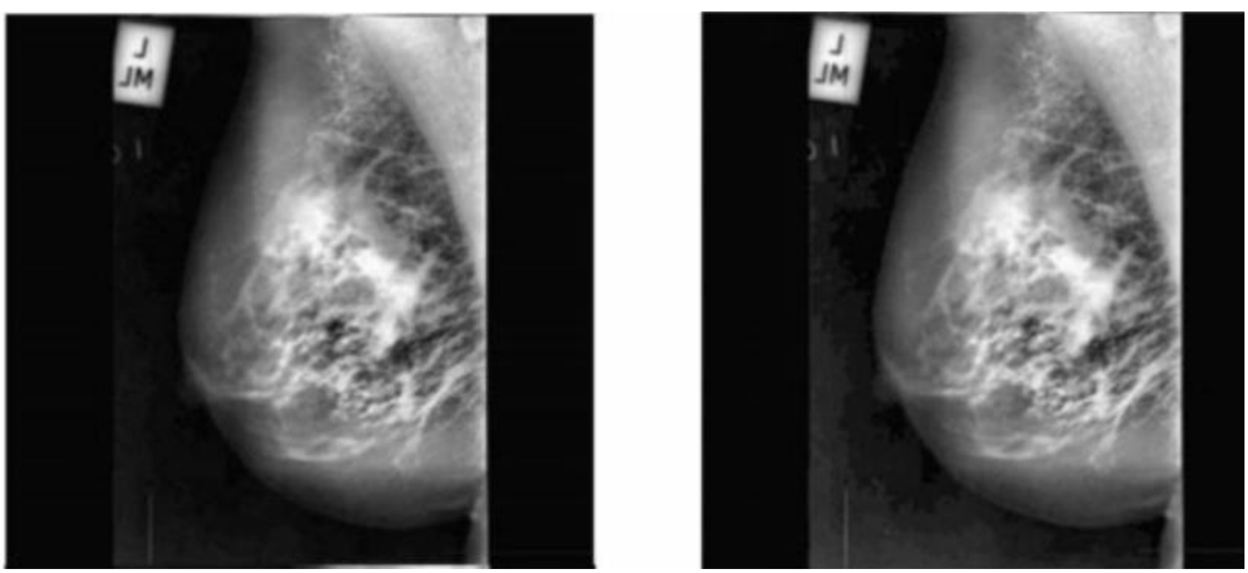


Figure 3: Image enhancement using CLAHE method

4.1.3 Segmentation

For the performance analysis three segmentation techniques fuzzy c-means clustering, k-means clustering and hybrid k-means algorithms will be used. Performance of each algorithm is measured with the aid of four parameters (sensitivity, accuracy, balanced accuracy and F-score).

4.1.3.1 K-means clustering algorithm

This segmentation technique partitions the image information into clusters called K where the algorithm repetitively goes through the below steps. It evaluates the mean of each cluster. Distance of data points one by one to the cluster's center from each cluster will be estimated. Finally each data point to the closest cluster will be allocated based on the distance calculated. After allocating data points, center of the cluster will be calculated again and it measures the new vector based on the new center. Below shows the steps need to be followed in the k-means algorithm (Aswathy & Jagannath 2020).

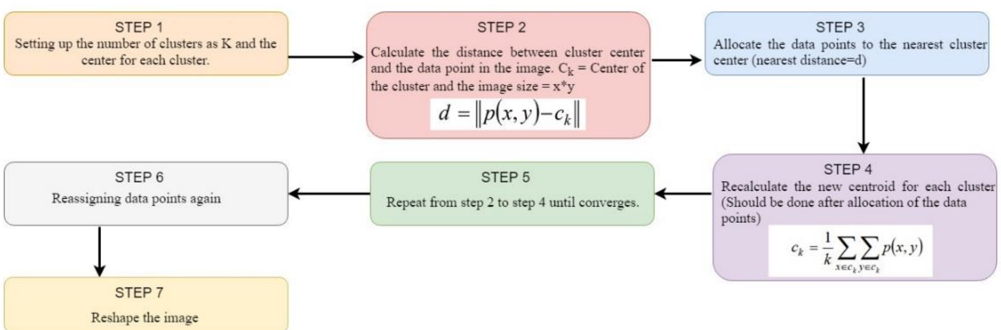


Figure 4: K means Clustering algorithm

4.1.3.2 Fuzzy c-means algorithm

Fuzzy c-means allocates membership to image's each data object which corresponds to centers of the cluster based on the distance between the data object and the center of the cluster (Ban et al 2016). Membership will be allocated if the data object is more near to the center of the cluster. After each and every iteration, the cluster center and the membership will be updated. Fuzzy C-means algorithm's classes are divided by centroids. The memberships are between 1 and 0 where each observation is classified into the degree of membership.

Below shows the equation which is built on reducing the cost function J_{fcm} (Aswathy & Jagannath 2020).

$$J_{fcm} = \sum_{k=1}^{n} \sum_{i=1}^{c} (u_{ik})^q \; d^2 (x_k, v_i)$$

$X = \{x_1, x_2, \ldots\ldots x_n\} \subseteq R^p$: Dataset vector population with p-dimension

n = overall sum of data

c= cluster number (range between 2 and n)

For the cluster i^{th}, u_{ik} = degree of membership

q = weighting function

v_i = i^{th} cluster center

$d^2 (x_k, v_i)$ = Euclidean distance

Below shows the steps of the algorithm in detail (Chattopadhyay, Pratihar & DeSarkar 2011).

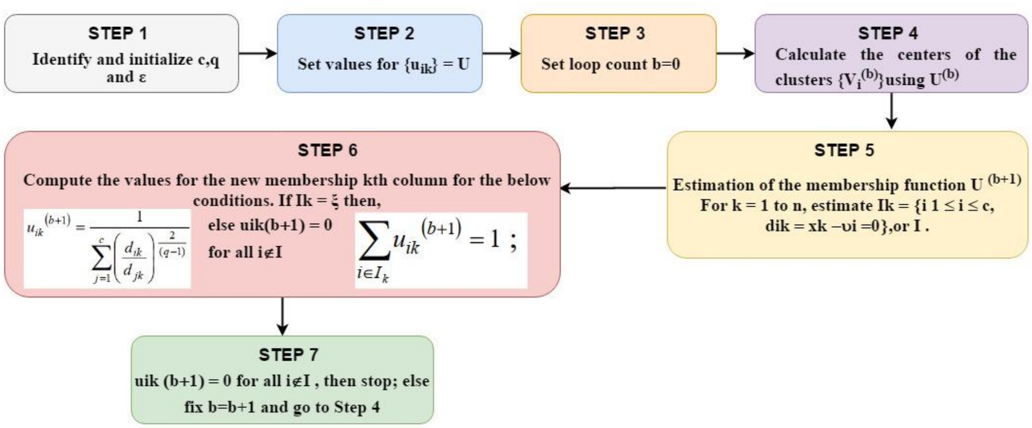


Figure 5: Fuzzy C means Clustering algorithm

4.1.3.3 Hybrid K-means algorithm

Hybrid K-means algorithm is built by modifying the traditional K-means algorithm where it produces better clusters comparing to the traditional K-means (Singh & Aggarwal 2014). Below diagram shows the hybrid K-means algorithm's steps in detail.

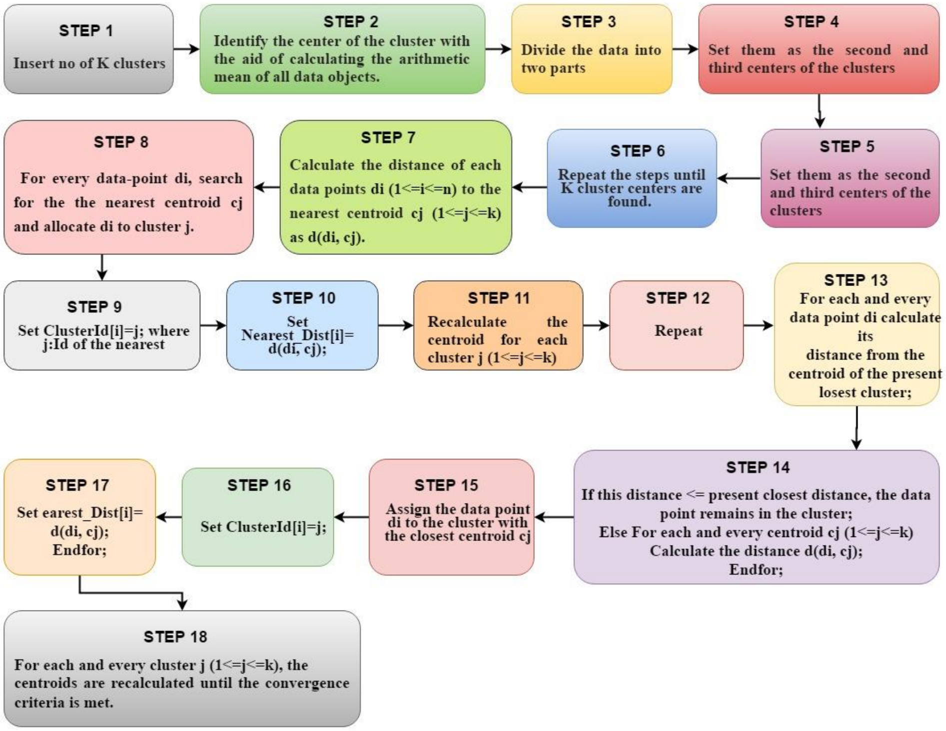


Figure 6: Hybrid K-means algorithm

4.1.4 Testing and Training

80% of the images will use as the training dataset while 20% will use for the testing purpose. Once the model is trained and tested, out of 58 images, 10 images were used for the cross validation to validate the results.

4.1.5 Performance evaluation quantitatively

Below shows the quantitative performance metrics which will to find the performance of three segmentation algorithms.

4.1.5.1 Accuracy

Accuracy compares the SI (segmented image) with the GTI (Ground Truth image) where it measures how accurately the image is segmented by the three algorithms (Mercan et al 2018).

True Positive (t_P): Cancer cell is correctly identified as a cancer cell (Parwar & Patil 2019)

True Negative (t_N): Normal cell is correctly identified as a normal cell

False Positive (f_P): Normal cell is incorrectly identified as a cancer cell

False Negative (f_N): Cancer cell is incorrectly identified as a normal cell

$$A = \frac{t_P + t_N}{N} \times 100$$

4.1.5.2 Sensitivity

Sensitivity represents the percentage of correctly segmented positive fraction (Aswathy & Jagannath 2020).

$$\alpha = \frac{t_P}{t_P + f_N}$$

4.1.5.3 Specificity

Specificity represents the percentage of correctly removed negative fraction (Aswathy & Jagannath 2020).

$$\beta = \frac{t_N}{t_N + f_P}$$

4.1.5.4 F-score

This is computed with the use of precision and recall functions where precision denotes the ratio between all the positives and true positives and recall to measure all the correctly identified true positives (Morstatter et al 2016).

$$\rho = \frac{t_P}{t_P + f_P} \quad \& \; \gamma = \frac{t_P}{t_P + f_N}$$

$$F - measure = 2 \times \frac{\rho \times \gamma}{\rho + \gamma}$$

4.1.5.5 Balanced Accuracy

Balanced accuracy measures the balance between two classes of sensitivity and specificity (Chambon et al 2018). Below shows the equations to calculate the above parameters.

$$B_a = \frac{\alpha + \beta}{2}$$

4.1.6 Analyze results

Once the results are noted from the above performance analysis, the results will be denoted with the aid of four graphs. First graph will show the three algorithms accuracy in the x axis and four parameters in the y axis while the rest illustrates the four performance measures in the x axis and the percentage of accuracy in the y axis.

4.2 Implementation

Implementation will be carried out in the Jupyter Notebook IDE, an open source software which uses for interactive computing across many programming languages (Jupyter 2020). Laptop of 1.99GHz processor and 16GB RAM will use in windows 10 platform. This will be implemented by using Java language. In addition, OpenCV a image processing library will be used (OpenCV 2020). It is a machine learning software library where it accelerates the usage of perception in machine in products where it consists of programming functions mainly aiming image processing techniques.

4.3 Limitations

One of the limitation in this research is that the model will be trained with only one data set which is consisting of 58 images considered as a less amount. The accuracy of the model will increase if the training image dataset is higher. Out of the segmentation techniques, for this research we will use only three algorithms. The research can be carried out with the use of variety of segmentation techniques so that the evaluation can done with a wider range of algorithms. Out of performance parameters, we will be only using four parameters for the evaluation where the performance can be analyzed with more parameters to analyze the best algorithm. In the preprocessing step, we will be using only one technique to enhance the image quality in the preprocessing phase.

4.4 Further work

This research will perform using UCSB dataset where the model can be trained with the aid of other datasets consisting more images such as Wisconsin dataset and UCI breast cancer dataset. It will enhances the accuracy since it allows various kind of test scenarios. Experimental results will be evaluated with the use of more performance parameters where only four parameters will be considered to obtain the best algorithm. Research can be developed by considering more number of clustering techniques where it will lead to observe more accurate segmentation algorithm. In this research we are focusing on unsupervised segmentation clustering techniques which are under region of interest segmentation algorithms. This research can be carried out by considering supervised segmentation techniques and also region based segmentation algorithms. To gain a high accuracy of the model, more preprocessing techniques can be used such as image smoothing and image enhancement techniques can be used.

5. Ethical Consideration

In any of the projects that are undertaken whether it be research or any other form, ethics plays a very crucial role in accomplishing that project. Ethics is a set of principles that are used for decision making as well as an agent for reconciling conflicting values (Anis 2017). Many times nowadays, it is heard about data breach cases all around us. Thus, ethics is the reflection of respect for all the end-users who are involved in the project. Moreover, it ensures all the participants in the project that no thoughtless, unsafe, or harmful demands are been made to them by researchers (Low and Johnson, 2014). Furthermore, it also ensures that the gained knowledge is been shared with all the ones that are involved in the research project (Anis, 2017).

The ethical issue which might take place in this research can be revealing of personal information of the patients or misusing them. The ICT involvement team includes a project manager, senior and junior software engineer, senior quality assurance engineer.

The research project that is initiated also has some ethics that would be taken into consideration which are as follows (ACS 2020):

- The code of ethics proposed by the Australian Computer Society is followed strictly (Michael 2016).
- Data gathered from internet resources are fully acknowledged.
- Data collected is always kept safe from unauthorized access.
- Data amassed is sufficient and not excessive.
- In no case, data collected is misused.

6. Project Management

This project is estimated to get delivered within 3 months. This project is divided into various phases and these phases are data gathering, image pre-processing, image Segmentation, train models, test models and finally analyzing the results.

6.1 Time

Below, Gantt chart is shows starting date and ending date of the project including starting and ending date of various phases involved in the project.

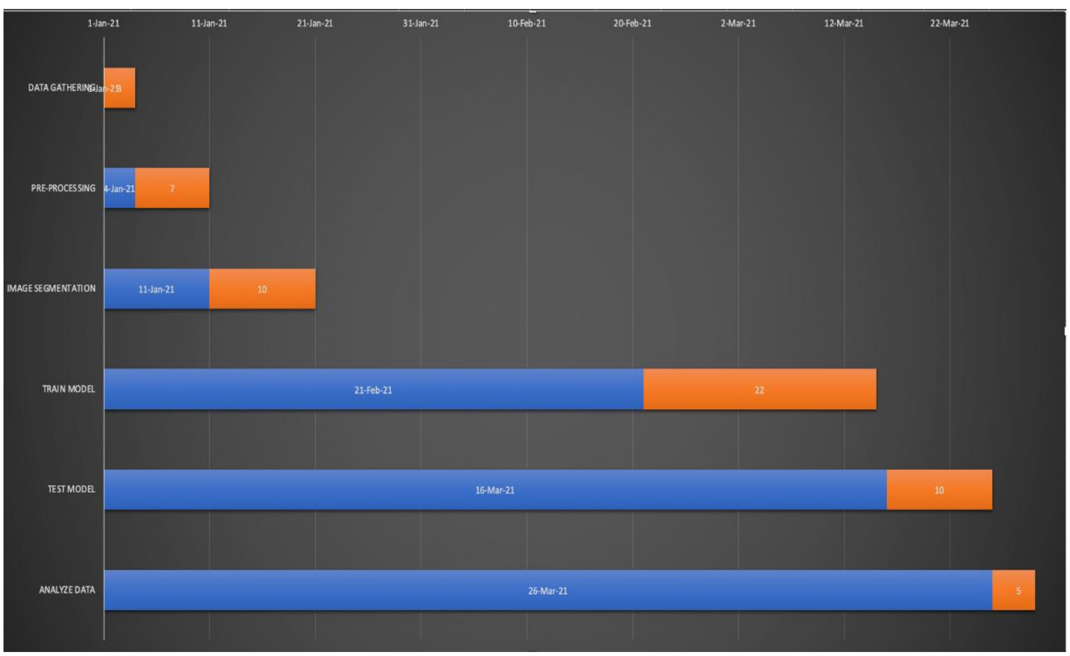


Figure 7: Gantt chart

Team will be supervised under an experienced project manager. Also, for successful completion of any project it is necessary that all team members maintains proper co-ordination (communication) among them. For achieving this, daily stand up meetings will be held. Online collaborative tool Jira will be used to assign and log their work weekly (Jira 2020). Apart, from using Jira, other tools like Email and google calendar will be used by team members for sharing other non-sensitive materials and communications.

6.2 Budget

Below table shows breakdown of the total budget of the project. Salaries of each member is evaluated (Glassdoor 2020).

Team Members	Salaries(in AUD) for 3 months	Total
Project Manager	32499	
Senior Software Engineer	29748	
Junior Software Engineer	17499	
Doctor(Cancer Specialist)	24999	
Senior QA Engineer	26748	
Total Cost for Staff		**131493**
Stationary	120	
Transportation (Meeting the doctor)	200	
Other cost		**320**
Total Cost		131,813

Table 1: Cost Evaluation

So, this research has been designed to run with the budget of **AUD 131,813$.**

6.3 Scope

In the research, three algorithms will be used by using four performance parameters with one data set (UCSB dataset). Data will be analyzed with the aid of five graphs.

6.4 Quality

Quality of the images will be enhanced with enhancement techniques and duplicate data will be not taken to account. Data accuracy will be assured. Black box and load testing will be carried out in the dataset to verify it has the acceptable output of F1 score. The data which are processed correctly need to be verified with the aid of white box testing. Module values can be tested by using unit testing.

6.5 Key Stakeholders

Key stakeholders who are interest in in this research are radiologists, pharmaceutical research companies, patients, medical students, oncologists, pathologists, specialists and nurses.

7. Risk Management Plan

Risk management plan will be created by the project manager of the team to predict the risks, estimate and response to impacts. Below describes the risk management plan for this research.

7.1 Strategies of Risk Management

Process Name	Process Description
Risk Identification	Project manager will conduct meetings before each phase of the project to discuss potential risks and new risks in the current project. Online tool will be used to identify and keep track of tools.
Risk Evaluation	In the end of each project phase, the risks are going to compare with the set of project's criteria where all the risks are ranked with the identification of priorities. If the risks are not meet the project criteria's they are disregarded.
Risk Prioritization	Identification and prioritization of risks according to the significance and importance
Risk Treatment	Project manager and the team members will come up with many options and solutions to address risks

Table 2: Strategies of Risk Management

7.2 Tools used in Risk Management

Risk Management Tools	Tool Description
SWOT Analysis	Strengths, Weaknesses, Opportunities and Threats are identified by the team
Risk Status Report	Weakly report is maintained to provide updates on actions and risks
Risk Register	Report will be maintained to handle the risks, responses to risks, and risk catogory for future references

Table 3: Tools used in risk management

7.3 Sources of Data

Data sources	Data sources Description
Work Breakdown Structure - Risk Report	Extracting from the WBS and includes recourses, time and cost
Cost Evaluation Risk report	Maintain throughout the project carefully since this report identifies risks relates to costs
Stakeholder analysis - risk report	Conflicts among key stakeholders should be tracked in the initial stage of the project

Table 4: Sources of data

7.4 Identification of Risks

Risk#	Description
RK001	Failure in servers, power outages or natural disasters
RK002	Data breach of sensitive data such as patient's images
RK003	Ransomware blocking devices and interrupting the implementation
RK004	Inaccurate results since using less number of images leading for inaccurate diagnosis of breast cancer
RK005	Reduction of speed of the internet connections
RK006	Less motivation in the team
RK007	Damages to electronic equipment i.e. laptops
RK008	Loss of physical assets which includes sensitive information such as patient's data
RK009	Tested samples can be destroyed and misplaced
RK010	Delay in implementation leads for increasing the allocated cost and budget
RK011	Backup data unavailable when they are needed
RK012	Software's prone to cyber attacks

Table 5: Identification of risks

7.5 Risk Matrix

	Severity -->		
Likelyhood	1	2	3
1	RK005		RK008 RK011
2	RK006	RK007 RK009 RK010	RK002 RK012
3		RK001	RK003 RK004

Low
Medium
High

Table 6: Risk Matrix

7.6 Contingency Plan

Scenario	Trigger	Response	Who to inform?	Key Responsibilities		Timeline	
				Who	What	What	When
The data collected may get loss mistakenly by any of the team member.	Instead of pressing save button, any member can press delete button.	Access backup database and get data immediately. Just make sure to perform timely backup	Project Manager, Developer	Project Manager, Developer	To see the problem and get data immediately	Alert the project manager and Developer	As soon as data is deleted or loss
The shutdown or unavailability of certain resource which is processing collected data.	Sudden powercut or resources hang up due to high computing	Access the standby resources and fetch data from the server and start working again	Project Manager	Project Manager	To make the team available with other standby resource temporarily	Alert the resource manager and project manager	As soon as the situation happen.

Table 7: Contingency Plan

Conclusion

Breast cancer can be considered as one of the major and dangerous disease among women in the modern society. Clinicians face issues in the detection since the diagnosis of histopathology images are subjective. The main goal of this research proposal is to identify the best algorithm out of selected segmentation algorithms namely fuzzy c-means algorithm, k-means, and hybrid k-means with the aid of four parameters named accuracy, f-score, sensitivity and specificity. Results will be represented with graphs to analyze the results easily. Paper presented the problem statement and research goals and aims initially. Proposed methodology uses UCSB data set containing 58 images with normal and abnormal (cancerous images) for the detection. Limitations of this research are training data set with limited number of images, using of one enhancement technique to enhance quality, considering only three clustering algorithms and four parameters for the detection. Future scope of this research can be applied on various histopathology images in different segmentation techniques for the identification of breast cancer. This research can be implemented and applied in real time breast cancer detection.

Word Count: 3508

References

ACS 2020, "*Code of Ethics 2020*", viewed 11 October 2020, <https://www.acs.org.au/content/dam/acs/acs-documents/Code-of-Ethics.pdf>

Amalia, AE, Airlangga, G & Thohari, ANA 2018, Breast Cancer Image Segmentation Using K-Means Clustering Based on GPU Cuda Parallel Computing, *Jurnal Infotel*, vol. *10, no.* 1 , pp. 33-38.

Anis, M 2017, "Research ethics," *SlideShare*, viewed 11 October, 2020, <https://www.slideshare.net/mobile/mohammeda24/research-ethics-78861141>

Aswathy, MA & Jagannath 2020, Performance Analysis of Segmentation Algorithms for the Detection of Breast Cancer, *Procedia Computer Science*, vol. 167, pp. 666-676.

Ataollahi, MR, Sharifi, J, Paknahad, MR & Paknahad, A 2015, 'Breast Cancer and associated factors: a review', *Journal of medicine and life*, vol. 8, no.4, pp. 6-11

Attia, J, Salim, S & Hussein, S 2017, 'Evaluation of Image Enhancement Techniques of Dental X-Ray Images', *Indian Journal of Science and Technology*, vol. 10, no. 40, pp. 1-5

Ban, OI, Ban, AI & Tuse, DA 2016, 'Importance–Performance Analysis by Fuzzy C-Means Algorithm', *Expert systems with applications*, vol. 50, pp. 9-16

Cancer Council 2020, *Types of Breast Cancer*, Cancer Council, viewed 20 October 2020, < https://www.cancer.org.au/cancer-information/types-of-cancer/breast-cancer >

Chambon, S, Galtier, MN, Arnal, PJ, Wainrib, G & Gramfort, A 2018, 'A Deep Learning Architecture for Temporal Sleep Stage Classification Using Multivariate and Multimodal Time Series', *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 26, no. 4, pp. 758-769

Chattopadhyay, S, Pratihar, Dk & DeSarkar, SC 2011, A comparative study of fuzzy c means algorithm and entropy-based fuzzy clustering algorithms, *Computing And Informatics*, vol. 30, no. 4, pp. 701-720

Dubey, Gupta & Jain 2016, Analysis of k-means clustering approach on the breast cancer Wisconsin dataset, I*nternational journal of computer assisted radiology and surgery*, vol. 11, no. 11, pp. 2033-2047.

Filipczuk, P, Kowal, M & Obuchowickz 2012, Breast fibroadenoma automatic detection using k-means based hybrid segmentation method, *In 2012 9th IEEE International Symposium on Biomedical Imaging (ISBI)*, Barcelona, pp. 1623-1626

Glassdoor 2020, *Junior Software Engineer Salary*, Glassdoor, viewed 20 October 2020, < https://www.glassdoor.com.au/Salaries/junior-software-engineer-salary-SRCH_KO0,24.htm>

Glassdoor 2020, *Junior Software Engineer Salary*, Glassdoor, viewed 20 October 2020, < https://www.glassdoor.com.au/Salaries/junior-software-engineer-salary-SRCH_KO0,24.htm>

Glassdoor 2020, *Melbourne Doctor Salary*, Glassdoor, viewed 20 October 2020, < https://www.glassdoor.com.au/Salaries/melbourne-doctor-salary-SRCH_IL.0,9_IM965_KO10,16.htm?countryRedirect=true>

Glassdoor 2020, *Melbourne IT Project Manager Salary*, Glassdoor, viewed 20 October 2020, < https://www.glassdoor.com.au/Salaries/melbourne-it-project-manager-salary-SRCH_IL.0,9_IM965_KO10,28.htm>

Glassdoor 2020, *Melbourne Senior QA Salary*, Glassdoor, viewed 20 October 2020, < https://www.glassdoor.com.au/Salaries/melbourne-senior-qa-salary-SRCH_IL.0,9_IM965_KO10,19.htm>

Glassdoor 2020, *Melbourne Senior Software Engineer Salary*, Glassdoor, viewed 20 October 2020, < https://www.glassdoor.com.au/Salaries/melbourne-senior-software-engineer-salary-SRCH_IL.0,9_IM965_KO10,34.htm>

Gowri, DS & Amudha, T 2014, 'A Review on Mammogram Image Enhancement Techniques for Breast Cancer Detection', *2014 International Conference on Intelligent Computing Applications*, IEEE Xplore All Conference Proceedings, ProQuest Central, pp. 47-51

Harbeck, N & Gnant, M 2016, 'Breast cancer', The Lancet, vol. 389, no. 10074, pp. 1134-1150

Heena, H, Om, P & Tripti, S 2014, 'Brain tumor segmentation: A performance analysis using K-Means, Fuzzy C-Means and Region growing algorithm', IEEE International Conference on Advanced Communications, pp.1621-1626

Janani, P, Premaladha, J & Ravichandran, KS 2015, Image Enhancement Techniques: A Study, Indian *Journal of Science and Technology*, vol. 8, no. 22, pp. 1-12

Jira 2002, *Jira Software,* Jira, viewed 25th October 2020, <https://www.atlassian.com/software/jira>

Juliet, F & Stacey, EL 2020, 'Genetic determinants of breast cancer risk', Current opinion in endocrine and metabolic research, Vol.15, pp.1-7

Jupyter 2020, *Documentation*, Jupyter, viewed 21 October 2020, <https://jupyter.org/documentation>

Kanungo, GK, Singh, N, Dash, J & Mishra, A 2015, 'Mammogram image segmentation using hybridization of fuzzy clustering and optimization algorithms', In Intelligent Computing Communication and Devices, vol.1, no.1, pp. 403-413

Kaur, N & Aggarwal, S 2017, 'Designing a New Hybrid K-Means Optimization Algorithm', *International Journal of Advanced Research in Computer Science*, vol. 8, no. 5, pp. 1567-1573

Kobashi, S, Nyul, L & Udupa, JK 2016, 'Soft Computing in Medical Image Processing', Computational and Mathematical Methods in Medicine, vol.2016

Kumar, KS, Venkatalakshmi, K & Karthikeyan, K 2016, ‘Performance investigation of assorted filters in medical image preprocessing for denoising using various noises', *Advances in Natural and Applied Sciences*, vol. 10, no. 11, p. 130

Leedy, PD & Ormrod, JE 2015, *Practical Research: Planning and Design*, Global Edition, 11th edn, Pearson Education Limited, Ebook Central (ProQuest).

Li, X & Plataniotis 2015, KN 'Color model comparative analysis for breast cancer diagnosis using H and E stained images', SPIE Medical Imaging, March 19, 2015, December 18, 2015, Orlando, Florida, United States, vol. 9420, viewed 20 October 2020, <https://doi-org.ezproxy.lib.swin.edu.au/10.1117/12.2079935>

Low, G & Johnson, M 2014, ACS Code of Professional Conduct Professional Standards Board Australian Computer Society

Mercan, C, Aksoy, S, Mercan, E, Shapiro, LG, Weaver, DL, Elmore, JG 2018, 'Multi-Instance Multi-Label Learning for Multi-Class Classification of Whole Slide Breast Histopathology Images', *IEEE transactions on medical imaging*, vol. 37, no. 1, pp. 316-325

Michael, B, Oliver, B, Don, G & John, W 2006, 'ICT Integrity: bringing the ACS code of ethics up to date', AJIS. Australasian journal of information systems, vol. 13, no. 2

Morstatter, F, Liang, W, Tahora, HN, Kathleen, MC, Huan, L 2016, 'A new approach to bot detection: Striking the balance between precision and recall', *2016 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, August 18 - August 21, 2016, San Francisco, CA, USA, viewed 20 October 2020, <https://ieeexplore-ieee-org.ezproxy.lib.swin.edu.au/document/7752287>

Najunath, BS 2019, *Bio-Segmentation*, The Regents of the University of California, viewed 21 October 2020, <https://bioimage.ucsb.edu/>

OpenCV 2020, *OpenCV is 20!*, OpenCV, viewed 20 October 2020, <https://opencv.org/>

Pawar, PS & Patil, DR 2019, 'Breast Cancer Detection Using Neural Network Models', 2013 International Conference on Communication Systems and Network Technologies, *IEEE*, April 6 - April 10, 2015, Gwalior, India, pp. 568-572, viewed 20 October 2020, <https://ieeexplore-ieee-org.ezproxy.lib.swin.edu.au/document/6524463>

Popovic, A, Fuente, M, Engelhardt, M & Radermacher, K 2007, 'Statistical validation metric for accuracy assessment in medical image segmentation', *International Journal of Computer Assisted Radiology and Surgery*, vol. 2, no. 3, pp. 169-181

Saunders, C, Jassal, S & Lim, E 2019, '*Breast Cancer*', Incorporated, 2nd edn, Oxford University Press, Ebook Central (Proquest)

Singh, P & Aggarwal, S 2014, 'A New Hybrid K-Means Clustering Algorithm', International Journal of Computer Science and Technology, vol. 5, no. 3, pp. 336-342

Stetco, A, Zeng, X & Keane, J 2015, 'Fuzzy C-means++: Fuzzy C-means with effective seeding initialization', *Expert systems with applications*, vol. 42, no. 21, pp. 7541-7548

Sun, Y, Zhao, Z, Yang, Z, Xu, F, Lu, HangJing, Z, ZhiYong, S, Jiang, J, Pingping, Z 2017, ‘Risk Factors and preventions 8 of Breast Cancer’, *International Journal of Biological Sciences*, vol.13, no. 11, p. 1387 – 1397

The Regents of the University of California 2020, *UC Santa Barbara*, The Regents of the University of California, viewed 21 October 2020, <https://www.ucsb.edu/>

Zheng, Yoon & Lam 2014, Breast cancer diagnosis based on feature extraction using a hybrid of K-means and support vector machine algorithms, *Expert Systems with Applications,* vol. 41, no. 4, pp. 1476-1478.